ELAN1 GROUP · TECHNICAL INVENTORY · v2.0

The full technical inventory

APIs, MCP, schemas, agents, memory, routes, tables and stack, across four products — revised for the Elan1 Govern build

PREPARED FOR	DATE	SCOPE	STATUS
M K Elango, founder	27 October 2026 (v1.0: 28 September 2026)	Elan1 Govern (formerly elan1) · rung1 · stack1 · verna1	Revised — draft for review

READ THIS FIRST

Four products, one kernel — and the kernel now has a name and a codebase. On 27 October 2026 the founder renamed the product **elan1** to **Elan1 Govern**; the legal entity stays **elan1 Private Limited**, and rung1, stack1 and verna1 keep their names. In v1.0 rows, read "elan1" meaning the product or kernel as Elan1 Govern. The corpus sentence still holds: *"rung1 is the product. stack1 is the runtime. elan1 is the kernel — the governed write path."* Signature rule unchanged: *a model may produce a write intent; it may never produce an allow* (E02 I1).

Status as at 27 Oct 2026. Elan1 Govern exists as working code — `github.com/Hasif-Elan1/elan1-govern`, commit `22388ad` — and is the **current, partial Level 1 implementation**: no meter, and enforcement only on its MCP route. Its scope is wider than the v1.0 kernel: it governs **every agent tool call, reads and writes, in any industry**, and adds a governance layer (AI inventory, assessments, 10 compliance frameworks). The kernel specification in section 3 remains the target; L2 is not claimed (two of its ingredients, Ed25519 signing and an offline verifier, exist in code); L3 is specified. There is no external customer. Tags: BUILT = present at 22388ad and confirmed by an independent read of the code (the demo was not run in the audit) · PARTIAL · SPECIFIED = frozen specification, no code · FIXED / REC / OPEN as in v1.0.

1 · THE FOUR PRODUCTS AT A GLANCE

Product	What it is	Status	Repositories
Elan1 Govern formerly elan1 · the kernel	Governance platform for AI agents: before any tool runs it checks a signed rulebook and answers allow · ask a person · refuse (default refuse); every decision is a signed, hash-chained receipt; governance layer on top. Specification: governed write path between an agent and a system of record — gates, human approval, single writer, hash-chained proof, meter; owns packs and the verifier.	PARTIAL L1 built in code (22388ad) · L2/L3 specified	**Hasif-Elan1/elan1-govern** (code of record, TypeScript monorepo) · **Ashoke-elan1/elan1-govern** (275-screen design) · elan1-core, elan1-declarations (+ 6 proposed)
rung1 the product	Governed AI agents for Indian SMEs (₹5–50 cr), sold via CAs. Lines: Enterprise (funded), Revenue, Commerce (parked), Voice (channel).	Enterprise funded · only priced product	rung1-enterprise (rung1-erp)
stack1 the runtime	Contract layer for the open-source AI stack: ten fusion.*/v1 contracts, 7-class conformance suite, 9-step substitution protocol with two human gates.	Live internally · GA Q4 2027	stack1-contracts, stack1-runtime
verna1 parked brand	Regulated outbound collections with a carrier partner. Reopens only on a paid NBFC pilot; own operating company from day one.	Parked (DR-007)	None active

2 · ELAN1 GOVERN — AS BUILT (commit 22388ad, 27 Oct 2026)

Area	Inventory
Code of record	One TypeScript monorepo (internal codename "gate", packages `@gate/*`, CLI `govern` = `gate`). 47 commits, 26 Oct 18:39 – 27 Oct 10:10 IST, one identity; ≈33,600 lines (≈31,300 TypeScript + Python). No licence file; no independent code or security review yet.
Workspaces (8)	`apps/gate` Fastify guard service (MCP + agent API, jobs, evidence) · `apps/web` Next.js console (27 pages, 53 API handlers) · `packages/core` pure logic (canonical JSON, crypto, rulebook, engine, receipts, verifier, bundle, detectors, suggestions, replay, assessments, frameworks, packs) · `packages/db` Prisma schema, tenant-scoped client, serialized chain writer · `packages/connectors` Gmail, Zoho Books · `packages/sdk` TypeScript SDK · `packages/sdk-python` stdlib-only Python SDK (`guarded` decorator) · `packages/cli` `govern verify · sign · keygen`.
Agent API (gate)	`POST /mcp` (MCP Streamable HTTP, stateless, official SDK) · `POST /v1/check` · `POST /v1/record` · `GET /v1/approvals/{id}` · `GET /v1/openapi.json` (OpenAPI 3.1, 5 paths, agent-facing only) · `GET /healthz`. Agent keys prefixed `gk_`, SHA-256 hashed, constant-time compare.
Two modes (renamed)	**Governed run** — MCP: Elan1 Govern holds the downstream credential and runs the tool (corpus: Enforced). **Check and record** — REST/SDK: Govern decides, the agent runs the tool with its own credentials and may skip Govern (corpus: Attested advisory). The code inverts v1.0's Mode A/B. Allowed and approved MCP calls run before their receipt is written (DECISIONS #15).
Decision path	authenticate agent → load signed rulebook (Ed25519 verified on every load) → resolve tool (built-in + synced upstream) → validate arguments → count limits → decide (pure, no I/O; strictest wins: refuse > ask > allow; no match = refuse) → run / hold / refuse → write receipt. System refusal ids (6): `agent-paused · no-rulebook · rulebook-invalid · default · invalid-arguments · gate-error`. Calls rejected before the guard (bad key, rate limit) produce no receipt.
Rulebook schema	YAML, signed per version (sign + publish, diff view). Rule = `tool` xor risk `tag`, conditions `when` (`eq lt lte gt gte in to_domain_in fields_include`), `unless` on ask rules, decision allow · ask (role/approver group) · refuse, flag "Can be approved in Slack or WhatsApp". Quotas: `max_count` / `max_sum` per `day` or rolling `Nm|Nh|Nd`. Try an action; Test on past actions (7/30 days, encrypted replay store, 30-day purge, switchable). Rule suggestions drafted nightly from repeated approvals.

Sources: E01–E27, O01–O12, Technology and Information Memoranda and Architecture Atlas (revised 27 Oct 2026), Four Doors v2.0, Elan1 Govern code audit (commit 22388ad), screen map, project notes. Draft — internal.

Area	Inventory
Risk tags (8)	`reads_data · writes_data · moves_money · sends_external · personal_data · deletes · changes_access · changes_production`; high-impact: moves_money, deletes, changes_access, changes_production.
Approvals	States pending · approved · rejected · expired (default 24 h, max 30 d; expiry = refuse). Channels web · email · phone link (signed, single-use) · Slack · WhatsApp. Password re-asked if sign-in > 10 min old; chat buttons skip it. Self-approval allowed but flagged when the approver created the agent. Emergency stop does not cancel approvals already waiting.
Receipt	22 top-level fields (strict zod): `seq ts org kind(decision|approval|outcome|admin) agent tool args_summary args_hash rulebook{version,hash} rule_id decision reason approval{…} outcome{…} ref_seq actor action details prev_hash kid hash sig`. Full arguments never stored. Hash = SHA-256 over custom JCS-like canonical JSON; Ed25519 (node:crypto) over the hash string; per-org chain from a zero genesis, `pg_advisory_xact_lock`.
Checkpoint · evidence	Hourly signed checkpoint of the chain head `{org, seq, hash, ts, kid}`. Bundle `gate-evidence-bundle/1`: receipts.jsonl, checkpoints.json, public_key.pem, public_keys.json, signed manifest.json, README; auditor pack adds a PDF and compliance map. `govern verify` fully offline, exit 0 intact · 1 broken · 2 usage; trusts the bundle key unless `--key <PEM>` is supplied.
Keys	Per-organisation Ed25519 signing key, wrapped by one environment master key (AES-256-GCM); rotation writes a `key.rotated` receipt; `KeyProvider` interface only — no KMS/HSM.
Console routes (27)	`/login /signup /onboarding /invite/[token] · / (Today) · /approvals /approvals/[id] /approve/[token] · /activity · /agents /agents/new /agents/[slug] · /rulebooks /rulebooks/[slug] · /connectors · /evidence · /inventory /inventory/[id] · /assessments /assessments/[id] · /compliance /compliance/[framework] · /alerts /alerts/[id] · /settings /settings/sso /settings/notifications`. 13 areas replace the Four Doors; 113 of 275 design screens sit on working code (38 one-to-one, 75 merged); 17 built screens have no design yet.
Console API (53)	`/api/` auth (login, signup, session, totp, sso start/callback) · agents (keys, status, behaviour) · approvals · activity · rulebooks (+ replay) · suggestions (apply, dismiss) · connectors · upstreams (sync, tools) · evidence (checkpoint, export) · receipts/[seq] · inventory · assessments (submit, review) · compliance (frameworks, profile, report) · alerts (+ settings) · emergency-stop (+ resume) · incidents/[id]/report · members · invites · orgs · settings (+ sso) · signing-key · webhooks (+ test) · notifications (slack, whatsapp) · demo.
Tables (30, Postgres 16, Prisma 6)	`User Session Membership Org Invite Agent ApiKey Rulebook Connector Approval ApprovalLink Receipt Checkpoint OrgFramework HealthAlert SsoConnection AiSystem Assessment UpstreamServer UpstreamTool OrgKey Webhook WebhookDelivery ReplayCall ChatChannel ChatNumber ChatMessage RuleSuggestion BehaviourAlert Incident`; enums `Role AgentStatus RiskTier ApprovalStatus`. Tenant isolation in application code (`forOrg()`) plus a test — **no row-level security**.
Connectors	Gmail, Zoho Books (customer's own OAuth app; mock by default; live mode never run against real accounts). "Your MCP servers": any upstream MCP server behind Govern, tools synced with suggested risk tags, switched on per tool, SSRF-protected; Govern holds the auth header.
Governance layer	AI inventory (name, kind, vendor, purpose, owner, unit, lifecycle, risk tier, data, affected people, impact, next review — no model-name field). Assessments: risk triage · impact assessment · data-protection check, sign-off by a different person. 10 frameworks (EU AI Act, NIST AI RMF, ISO 42001, DPDP, RBI FREE-AI, GDPR, HIPAA, SOC 2, Colorado AI Act, NYC LL 144), 13 controls GATE-01…13, coverage counted from receipts; all "Draft — needs legal review". 7 industry packs: finance, healthcare, HR, support, sales, engineering, general.
Alerts · stop	6 detectors: volume_spike, refusal_surge, amount_outlier, new_counterparty, new_tool, off_hours. Auto-pause off by default, high severity only (3 detectors can reach high). Emergency stop: owners/admins, password + typing STOP, pauses every agent, writes an incident report.
Identity · roles	argon2id passwords (≥ 12 chars), TOTP, lockout; OIDC SSO (Okta, Microsoft, Google) with DNS-verified domains, SSO-only option, JIT users; single-use invites. 5 roles Owner · Admin · Approver · Viewer · Auditor, 22 permissions; auditors switch between client organisations.
Jobs · webhooks	pg-boss on Postgres: approvals-expire (1 min) · checkpoints (hourly) · webhook-sweep (1 min) · replay-purge (hourly) · rule-suggestions (nightly) · behaviour-scan (5 min); queues webhook-deliver, chat. Webhooks HMAC-signed, up to 8 delivery attempts, no outbox. Rate limits and tool-registry cache per process; multi-instance untested.
Tests (audit, 27 Oct)	Unit: 397 of 398 passed (1 needs `.env`). 41 test files: 21 passed, 20 failed (19 could not load the database client — Prisma download blocked; 1 `.env`). Python SDK 40/40. 23 Playwright tests in 9 specs, not run. Not product measurements.

Sources: E01–E27, O01–O12, Technology and Information Memoranda and Architecture Atlas (revised 27 Oct 2026), Four Doors v2.0, Elan1 Govern code audit (commit 22388ad), screen map, project notes. Draft — internal.

3 · ELAN1 GOVERN — THE KERNEL SPECIFICATION (v1.0 content, with status now)

Contract	Inventory (v1.0, unchanged)	Now — at 22388ad
KernelPort API	6 calls, identical for every consumer: define · assert · seal · verify · scope · refuse. 5 objects on the port: write intent, decision, approval, record, meter event. 8 invariants (I1–I8). Python protocol in elan1_port — the only import a consumer may use.	SPECIFIED Code exposes check · record · approval-status instead.
Right Path	9 rungs + 0b, fixed order, no fast path / admin mode / decision cache: 0b STOP · 1 IDENTITY · 2 WRITE DOOR · 3 PRE-GATES · 4 evaluate() · 5 GUARDRAIL · 6 AUTONOMY · 7 ACT · 8 PROOF · 9 METERED. Implemented as stages S01–S12 over an append-only WriteContext.	PARTIAL Stop, identity, evaluate, act, proof built; no write door, autonomy or meter.
Gates	G1 consent · G2 residency · G3 retention · G4 dual_control · G5 rate · G6 budget · G7 counterparty · G8 schedule · G9 precondition. Six can never be skipped (G1–G4, G7, G9). All run even after first failure.	SPECIFIED Code has rule conditions + quotas (≈ rate, budget); no dual control.
Autonomy	A0 Observe · A1 Draft · A2 Propose · A3 Act reversible · A4 Act bounded · A5 Act broad. Missing grant = A0. An agent may never approve; approvals are human, named, expire (4 h default) and re-climb every rung.	SPECIFIED Built: per-tool allow/ask/refuse; human approvals, 24 h default.
Blast radius	one_record · one_counterparty · batch · tenant · cross_boundary — also the normative meter basis.	SPECIFIED
Outcomes	committed · refused · escalated (synchronous). A refusal is HTTP 200, never 4xx; 429 + Retry-After for throttling.	BUILT as allow · ask a person · refuse.
Refusal register	43 typed codes (policy 14 · gate 19 · platform 10), 28 active at L1. Wire form POLICY.* / GATE.* / PLATFORM.*, e.g. POLICY.APPROVAL_DECLINED, GATE.SANDBOX_DIVERGED, PLATFORM.CHAIN_HEAD_CONFLICT.	SPECIFIED Code: 6 system rule ids + customer rule ids.
Tokens	Three classes, no fourth: tenant (one tenant, one actor, one autonomy level; the only class reaching /v1/writes), control (Studio, registry), ops (read-only, no tenant data). Plus a stack1 service token (O03).	PARTIAL Agent keys (gk_) + console sessions.
SURFACES · MCP · APIS · ROUTES		
Two modes	Mode A · Enforced — in-process SDK; the kernel holds the target credential (rung1 hosted). Mode B · Attested advisory — MCP or HTTP; "evaluated and recorded", never called blocking.	BUILT inverted: governed run (MCP, enforced) · check and record (SDK/REST, advisory).
MCP server	34 tools in 3 scopes: 18 ro (cannot reach rung 7), 10 stage (e.g. stage.write — validated request, no side effects), 6 act (e.g. act.write). Stateless; descriptions pinned by hash in the declaration; official MCP SDK, 2026-07-28 revision (REC); registered in stack1 fusion.tools/v1.	PARTIAL `/mcp` exists (stateless, official SDK); tools = built-in + upstream, not the 34-tool set.
HTTP API /v1	8 families in 3 classes — act: /v1/writes, /v1/obligations · ro: /v1/records, /v1/evidence (paged by seq), /v1/meter · ctl: /v1/declarations, /v1/actors, /v1/tenants. OpenAPI 3.1 (REC).	PARTIAL /v1/check, /v1/record, /v1/approvals/{id}; OpenAPI 3.1 published.
SDKs	Mode A: Python elan1_port (rung1 on Frappe). Mode B: Python + TypeScript @elan1/kernel; client-generated intent_id; retry 5xx/429 max 5, never a refusal.	BUILT @gate/sdk (TS), Python `guarded`; no idempotency key.
Webhooks	HMAC-signed (Standard Webhooks), at-least-once, carry seq + digest for gap detection; obligation.resolved also sent as a fusion.run/v1 signal.	BUILT HMAC-signed, retries; no outbox.
Console routes	Four Doors only — Work · Records · Agents · Assurance, never a fifth. elan1 Console screens: Studio (author→certify), Decide, Assurance, Meter. Doors appear in 4 consoles: elan1 Console, rung1 Console, rung1 CA console, stack1 P4 AI Governance. v3 screen set = 275 platform + 29 reference screens.	Superseded for Elan1 Govern by 13 areas, 27 routes (section 2).
Website routes	Astro static site elan1-web: /product/*, /assurance/*, /verifier, /action-receipt, /solutions/*, /obligations/* (MCA, GST, DPDP, RBI, EU AI Act, ISO 42001/NIST), /developers/* (quickstart, sdk, mcp, api, conformance, versioning), /refusal-codes, /glossary, /trust, /pricing, /pilot, /book, llms.txt, sitemap.xml.	SPECIFIED No site in the repo.
SCHEMAS · DECLARATION · PROOF RECORD		
Declaration	Policy as data (JSON, < 900 lines). JSON Schema 2020-12, additionalProperties:false (FIXED). 7 blocks: identity · shape · permissions · write_types · gates · proof_binding · channels. 11 semantic rules (L0-S01–S11) + 7 lint rules, run in CI with no platform. Pipeline: author → validate → simulate → certify (a governed write, dual control) → compile → resolve. Explicit integer version, never "latest". Validators: jsonschema (Py), Ajv (TS).	SPECIFIED Built counterpart: signed YAML rulebook (section 2).
Ontology (L3)	Controls and Regulations families complete the ontology; five-link chain clause → control → gate → write type → proof field (E21).	SPECIFIED Built: framework → control → receipt count.

Sources: E01–E27, O01–O12, Technology and Information Memoranda and Architecture Atlas (revised 27 Oct 2026), Four Doors v2.0, Elan1 Govern code audit (commit 22388ad), screen map, project notes. Draft — internal.

Proof record	16 frozen fields: F01 tenant_id · F02 seq · F03 prev_digest · F04 write_id · F05 write_type · F06 declaration_id · F07 declaration_version · F08 policy_version · F09 actor_id · F10 actor_kind · F11 autonomy_level · F12 record_id · F13 record_version · F14 payload_hash (salted) · F15 lexicon_hash · F16 recorded_at. SHA-256 over a length-prefixed preimage; no JSON in the digest path; unsigned at L1.	SPECIFIED Built receipt differs: 22 fields, canonical JSON, Ed25519-signed. Two formats — OPEN.
Meter event	event_id, write_id, seq, basis, quantity, outcome. DB CHECK: refused ⇒ quantity 0. Nightly two-way reconciliation with the proof log. Billable unit: the Governed Action.	SPECIFIED No meter in code.
Pack template	11 blocks; 5 admission gates (G-A dated hook · G-B named auditor · G-C reversibility · G-D evidence sold separately · G-E elan1 owns the write).	SPECIFIED Built industry packs (7) are starter configs, not admitted packs.

DATA STORES AND TABLES (SPECIFIED: PostgreSQL 16+, RLS on every table, ap-south-1 Mumbai, 3 AZ) — as built see section 2

Store	Tables	Retention / rule
Declaration registry	declaration · write_type · certification · surface_plan · policy_rule · lexicon · lexicon_term	Insert-only, kept for ever
Action store	write (intent/decision) · gate_result · stop_event · idempotency index (30 d) · outbox	Insert-only
Approvals	obligation · escalation · approval (with rendered-payload digest) · autonomy_grant	Each resolution is a governed write
Proof log	seal (16 fields, UNIQUE tenant_id+seq); INSERT to chain writer only, UPDATE/DELETE to none	Indefinite / 7 y / ≥ 8 y — conflict
Meter ledger	meter_event · meter_basis	As the chain
Records & snapshots	record_version · snapshot refs (30-day rollback) · webhook delivery log	30 days (snapshots)
Keys, salts, anchors (L2)	key · key_window · erasure_salt · anchor_batch · anchor_token · inclusion_path · bundle	Anchors indefinite
Pack & profile registry	portfolio register · verifier_profile · pack artefacts	Append-only

COMPONENTS, REPOSITORIES, PLANES

Services	e1-write-api · e1-kernel (stages) · gate engine · e1-policy (own rules evaluator; Cedar fallback) · e1-registry + compiler · e1-lexicon · obligation/approval · stop service · e1-connect + e1-keys (single writer, credential broker: single-use 10-min credential) · e1-seal + e1-evidence (chain writer) · e1-meter · MCP server · HTTP API · SDKs · webhook dispatcher · renderers + console · integrity jobs · reference tenant + conformance · pack/profile registry · L2 anchor/bundle/key/salt · elan1-verify.	PARTIAL Built as 2 apps + 6 packages (section 2); single chain writer per org.
Internal interfaces	KernelPort · Stage (continue/refuse/escalate) · Outbox → chain writer (pg_advisory_xact_lock, FOR UPDATE SKIP LOCKED) · Registry.resolve · PolicyEvaluator.evaluate · CredentialBroker.mint · TargetAdapter (snapshot, restore, apply → inverse, recheck).	PARTIAL Advisory-lock chain writer built; no outbox, no adapters.
Repositories	Recorded: elan1-core, elan1-declarations. Proposed: elan1-packs (config only, no src/), elan1-conformance (C1–C14), elan1-console, elan1-spec (64 vectors), elan1-verify (Go stdlib, empty lockfile), elan1-seal-b (TypeScript).	Code lives in Hasif-Elan1/elan1-govern; repo structure — OPEN.
Planes / zones	P0 residency & tenancy · P1 runtime (stack1) · P2 system of record · P3 kernel (shut) · P4 declaration · P5 surfaces · P6 channels · P7 apps · P8 packs. Zones Z1 edge → Z5 offline verifier.	SPECIFIED
Conformance & CI	C1–C14 consumer tests (C14 = claim-boundary copy lint). import-linter, dependency-cruiser, depguard; "no model reachable from rungs 3–8" (openai/anthropic/litellm forbidden); no-screen-files glob; licence check; nightly chain verify + write↔record join + meter reconcile + determinism replay.	PARTIAL Vitest, Playwright, semgrep, gitleaks, ZAP, pnpm audit configured; CI history not seen.

EVIDENCE AND CRYPTOGRAPHY (L2 Attested · L3 Proven — specified, not built)

L2	Seal Spec v6: pure Ed25519 (RFC 8032) over the frozen preimage · 64 test vectors (12 structure, 12 chain, 12 Merkle/anchor, 28 negative) · two independent implementations (Python/PyNaCl + TypeScript/node:crypto) · Merkle batching RFC 6962 · two RFC 3161 TSAs (eMudhra + one global), ≤ 15 min · 8-member evidence bundle · offline verifier in Go, 6 exit codes · verifier profiles · AWS KMS Ed25519 per residency domain · S3 Object Lock WORM · SD-JWT (RFC 9901) redaction · optional SCITT transparency log.	SPECIFIED L2 not claimed. In code: Ed25519 signing (one TS implementation), offline verifier (3 exit codes), checkpoints; no TSA, KMS, vectors, profiles.
L3	Cross-organisation verification · selective disclosure · precedent & earned autonomy · customer-held keys (CloudHSM / Luna / nShield) · sovereign deployment · Action Receipt open standard. Entry only after an external auditor accepts an L2 bundle in writing (Oct 2029 earliest).	SPECIFIED
Packs	ledger1 (MCA audit trail, first, Q1 2028) · invoice1 (GST e-invoicing) · collect1 (RBI collections, sketch). Reserved: manufacture1, realestate1, health1, bank1, insure1, telco1, gov1.	SPECIFIED Regulatory packs; distinct from the 7 built industry packs.

Sources: E01–E27, O01–O12, Technology and Information Memoranda and Architecture Atlas (revised 27 Oct 2026), Four Doors v2.0, Elan1 Govern code audit (commit 22388ad), screen map, project notes. Draft — internal.

4 · RUNG1 — THE PRODUCT (FIRST CONSUMER)

Product lines	Enterprise — ERPNext/Frappe, hosted mode, ₹2,499 / 7,999 / 19,999 / 49,999+ per month (funded; attach mode + Tally migration Phase 2) · Revenue — 6 CRM workflows, Twenty then Zoho CRM (parked; opens Q3 2027) · Commerce — Shopify, WooCommerce, ONDC, gates GC1–GC6 (parked) · Voice — inbound-only approval channel on P6 (pilot Q4 2027).
Kernel use	Mode A in-process SDK, Enforced; builds no gates, approvals, proof or meter of its own (DR-001). Pays elan1 ₹0.40 per committed action from 1 Jan 2027; refusals ₹0. **Now:** Elan1 Govern has no in-process enforced mode and no ERPNext connector; how rung1 consumes it is OPEN.
Systems of record / adapters	ERPNext via Frappe's own API (never the ORM); adapter returns the inverse op for 30-day rollback. Tally: read-only migration source. Linked, not copied: Zoho Books, Odoo, Twenty, HubSpot, Salesforce.
Schemas / write types	Declaration e.g. erp.ar v7, write type erp.sales_invoice.submit; channel rung1_mcp (kind mcp).
Agents	"rung1 drafting agent" registered as actor (actor_kind agent); actor_id = requesting human via on_behalf_of (R04). A0–A5 grant per write type.
Channels	Web console, WhatsApp (approvals — which write types is open, Q21), rung1 Voice, e-mail, API, MCP. Channels route into a door, never a door.
Consoles	rung1 Console (one per tenant) and rung1 CA console (multi-client queue, one audit export per client). Enterprise/Revenue/Commerce/Voice are filters inside the Four Doors.
Voice stack	LiveKit (framework only), Pipecat, Silero VAD, Smart Turn; LiveKit turn-detector weights BLOCKED. Read-back before every decision; no dial-out ever.
Repos / docs	rung1-enterprise (rung1-erp), package rung1_core (may not import elan1_kernel). R01–R24 series.

5 · STACK1 — THE RUNTIME

Contracts	Ten fusion.*/v1 contracts; five ship in Phase 1 to rung1 and elan1 only: gateway (C2 · LiteLLM), trace (C6 · Langfuse core, OTel/OpenInference, 90-day), tools (C7 · ContextForge MCP registry, per-call authz, call ledger), run (C8 · Temporal; Prefect/Dagster substitutes), guard (C10 · Presidio, NeMo — PII, provenance, injection defence). Also named: inference (vLLM) and memory (limited availability; GA 1 Oct 2027). The remaining contracts are defined in the S-series, not in this folder. **Now:** Elan1 Govern runs standalone (Postgres + pg-boss), not on fusion.*/v1 — OPEN.
Profiles	P1 Private Assistant · P2 Document Intelligence · P3 Agent Automation · P4 elan1 Layer Host / AI Governance (the only stack1 surface with the Four Doors).
Not substitutable	fusion-edge ingress (mTLS, signed tenancy assertion, attribution triple) · secrets (OpenBao, one path per tenant) · metering plumbing (same-transaction outbox) · deploy pipeline (Promoter, inverse.json) · sandbox pool.
Infrastructure	Managed Kubernetes ap-south-1, namespace per zone, default-deny · Postgres 16+ with RLS + synchronous standby · Keycloak (realm per tenant) · OpenTofu, Helm, Argo CD · OTel, Prometheus/Mimir, Loki, Tempo, Grafana · Object Lock storage (Garage/Ceph/SeaweedFS in sovereign mode).
Terms & rules	99.5% monthly per contract · contract p95 < 40 ms · meter lag p99 < 30 s · change windows Tue/Thu 21:00–23:00 IST. Boundary rules B1–B8: stack1 never imports elan1; reserved words approve/allow/gate/refuse/proof/seal; no UPDATE/DELETE on evidence.
Docs	S01–S19. GA: all contracts blocking-green 30 Sep 2027 (S13 says nine, memory LA — conflict A188).

6 · VERNA1 — PARKED

Scope	Regulated outbound collections with a carrier partner; would be its own plane. Reached through a CarrierPort. Shares no code path with rung1 Voice.
Blockers	TCCCPR and DLT registration, 1600-series calling, national preference register, consent state. Name collides with Skit.ai (formerly Vernacular.ai).
Tech assets	A 16-code typed refusal register (reused as the pattern for the kernel's). No active repository, stack or roadmap.
Reopen	Paid NBFC pilot only: signed order, named carrier partner, money received — then its own brand, own opco, cleared name.

7 · AGENTS, MODELS AND MEMORY (CROSS-CUTTING)

Agents	Agents are actors (actor_kind human · agent · system, set by the kernel from the token). Autonomy A0–A5; earned autonomy from precedent at L3. Agents never approve and never sit on rungs 3–8. **Now:** Elan1 Govern agents have an owner, risk level, keys, rulebook and instant pause; added by a 3-step wizard ending in an MCP address and key for Claude, Cursor, LangChain and others.
Models	Self-hosted open-weight models on vLLM in Mumbai (hosted provider only if on the sub-processor list). Reached via fusion.gateway/v1, upstream of assert — never on the decision path. **Now:** no model on the Elan1 Govern decision path (decide is pure code).
Memory	Runtime: fusion.memory/v1 (stack1). Semantic: lexicon term sets sealed as F15. Institutional: declarations kept verbatim for ever; precedent store (L3). Vector stores and caches partitioned by tenant. **Now:** Elan1 Govern keeps per-agent behaviour baselines (alerts) and an encrypted 30-day replay store.

Sources: E01–E27, O01–O12, Technology and Information Memoranda and Architecture Atlas (revised 27 Oct 2026), Four Doors v2.0, Elan1 Govern code audit (commit 22388ad), screen map, project notes. Draft — internal.

Build harness	Claude Code: CLAUDE.md, subagents, commands, hooks, skills (FIXED in the Build Book). Your installed skills include adr-scribe, census-derive, ci-green-loop, governed-write-review, k4-skill-author, pack-compose, r1-scout, r3-mutation-proof, sdk-boundary-check, vault-sync (Obsidian vault) and others.
Prototype	elan1-platform tree: FastAPI enterprise1 API (:8015), React/Vite UI (:5189), Postgres + Redis, Supabase (dev only), sor_write.governed() chokepoint, 13-field sealed tuple, 4 divergent declaration dialects. Writes only to its own Postgres, not an external system of record yet. **Now:** superseded as the lead implementation by Elan1 Govern (section 2); what carries forward is OPEN.

SHARED TECHNOLOGY STACK — SPECIFIED (v1.0) AND AS BUILT

Layer	Specified (v1.0)	Elan1 Govern as built (22388ad)
Kernel	Python 3.12, FastAPI, uv (REC; Go if Gate 1 misses)	TypeScript, Node 22, Fastify, tsx
Durable execution	Temporal via fusion.run/v1 (FIXED)	pg-boss (Postgres) — conflicts with DR-002, OPEN
Database	PostgreSQL 16+, RLS, CHECK, grants (FIXED)	PostgreSQL 16, Prisma 6; app-level tenancy, no RLS
Queue	Postgres outbox, no broker (REC)	pg-boss queues; no outbox
Console / web	React 18, Vite, TS, Tailwind; Astro site	Next.js 15.5, React 19, Tailwind, Radix; no site
Identity	Keycloak + own kernel token issuer	Own: argon2id, TOTP, OIDC (openid-client)
Secrets	OpenBao per-tenant paths (INHERITED)	Env master key; AES-256-GCM-wrapped org keys
Observability	OpenTelemetry, Grafana stack, Sentry	pino logs, /healthz, health alerts
Delivery	GitHub Actions, OpenTofu, Helm, Argo CD	Dockerfile, docker-compose (untested)
Supply chain	cosign, SLSA, Syft, Grype/Trivy, CodeQL, Semgrep, gitleaks	pnpm audit, semgrep, gitleaks, ZAP baseline (scripts)
Tests	pytest+Hypothesis, Vitest, Playwright, pgTAP, Testcontainers	Vitest + fast-check, Playwright, Python unittest
Alerting	healthchecks.io; paging vendor OPEN	In-app alerts, Slack/WhatsApp, signed webhooks
Other libraries	—	zod 4, yaml, fflate, pdf-lib, nodemailer, @modelcontextprotocol/sdk

8 · WHAT WAS MISSING FROM THE LIST — AND WHAT TO ADD

Also in the inventory	Metering and billing pipeline (meter → rating → invoicing; internal transfer billing) · security baseline (O05: India residency, 180-day logs, CERT-In, DPDP erasure via salt destruction) · open-source licence register (O04: 25 rows, none yet confirmed by counsel; LiteLLM/Langfuse ee/ excluded; Windmill, n8n, Inngest BLOCKED) · versioning policy (O06: breaking changes get 12 months) · SLOs, DR (no second region without a written decision), runbooks, status page. **Now:** none of these is built in Elan1 Govern.
Open conflicts to close	Proof retention (7 y vs ≥ 8 y MCA vs indefinite) · L2 date (Q3 2027 vs Apr 2028) · kernel latency (64 vs 91 vs 120 ms p50, all unmeasured) · hosting vendor (OD-28) · F14 salting scheme (due 30 Nov 2026) · legal entity (one or three) · trade mark clearance for rung1 and stack1. **New (27 Oct):** specification vs build as the target · two record formats (Seal v6 vs built receipt) · pg-boss vs Temporal (DR-002) · TypeScript vs the specified Python/Go stack · billing unit and read metering · rung1's path to Elan1 Govern (no in-process mode, no ERPNext connector) · Elan1 Govern on stack1 or standalone · trade mark clearance for "Elan1 Govern" · code ownership and IP assignment (no licence file) · independent code and security review.
Recommended additions	OpenAPI 3.1 spec and MCP tool catalogue as published artefacts · a webhook event catalogue · an agent/actor registry schema · threat model and DPIA per product · SBOM per release · ADR register and C4 diagrams per product · an error-budget policy · data dictionary for every table above · rung1 adapter contract tests per system of record · a stack1 contract list (all ten, with IDs) in one place. **Progress:** OpenAPI 3.1 now exists for the agent API (5 paths) PARTIAL ; an agent registry exists in code BUILT ; Elan1 Govern's DECISIONS.md and SECURITY.md are a start on an ADR register and threat model PARTIAL . **Add:** data dictionary for the 30 built tables · RLS · KMS/HSM key custody · an outbox for webhooks · multi-instance rate limiting · run the database and Playwright suites in CI.

Every figure above is a specification or target unless marked measured. Designed is not shipped (DR-010) — and built is not shipped to a customer. Companion documents: Elan1 Govern Information Memorandum, Technology Memorandum and Architecture Atlas, revised editions of 27 October 2026.